Project 1

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Problem 1

- 365 days, population fixed $N > 0$, state space S = S, I, R = 0,1,2

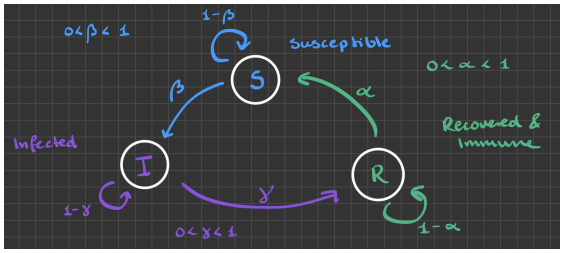


Figure 1: Transition Diagram

a)

- we look at one specific individual, Xn its state at time n, with X0 = 0. - the exercise states that each day, all individuals in a fixed state has the same probabilities fro moving or staying, meaning these probabilities must be independent of the individuals "history", i.e. how they ended up in this state. Thus, the markov property $P(X_{t+1} = j|X_0 = i_0, X_1 = i_1, ..., X_t = i) = P(X_{t+1} = j|X_t = i)$ is satisfied, and $P(X_{t+1} = j|X_t = i) = P_{i,j}$ where $P_{i,j}$ are the elements in the transition probebility matrix $\mathbf{P}$.

We index the matrix with rows $0, 1, 2$ and columns $0, 1, 2$ as such:

$$\mathbf{P} = \begin{bmatrix} P_{0,0} & P_{0,1} & P_{0,2} \\ P_{1,0} & P_{1,1} & P_{1,2} \\ P_{2,0} & P_{2,1} & P_{2,2} \end{bmatrix},$$

where the indices i correspond to "from" and j correspond to "to".

We were given $P_{0,1} = \beta$, $P_{1,2} = \gamma$, $P_{2,0} = \alpha$, and we know that $P_{0,2} = P_{1,0} = P_{2,1} = 0$.

From a given state i, the individual must either remain in i or move to state j with probability $P_{i,j}$, meaning the probabilities must satisfy $P_{i,i} + P_{i,j} + 0 = 1$.

Thus, $P_{0,0} = 1 - \beta$, $P_{1,1} = 1 - \gamma$ and $P_{2,2} = 1 - \alpha$.

We fill in the matrix and get $\mathbf{P} = \begin{bmatrix} 1-\beta & \beta & 0 \\ 0 & 1-\gamma & \gamma \\ \alpha & 0 & 1-\alpha \end{bmatrix}$.

b)

Given $\beta = 0.01$,$\gamma = 0.10$ and $\alpha = 0.005$. We check if the matrix is regular: $\mathbf{P}^2 = \begin{bmatrix} 0.99^2 & 0.99 \cdot 0.01 + 0.9 \cdot 0.01 & 0.01 \cdot 0.10 \\ 0.10 \cdot 0.005 & 0.9^2 & 0.9 \cdot 0.10 + 0.10 \cdot 0.995 \\ 0.005 \cdot 0.99 + 0.995 \cdot 0.05 & 0.01 \cdot 0.005 & 0.995^2 \end{bmatrix}$.

As we can see, all elements $P^2_{i,j} > 0$, and the matrix $\mathbf{P}$ is regular. From Thm. 13 in Ellings presentations, we them have guaranteed existence of a unique limiting distribution satisfying $\vec{\pi} = \mathbf{P}^{\mathrm{T}}\vec{\pi}$ and $\vec{1}^{\mathrm{T}}\vec{\pi} = 1$.

We solve for $\vec{\pi}$ and get $\vec{\pi} = \begin{bmatrix} 10/31 \\ 1/31 \\ 20/31 \end{bmatrix}$.

Thm. 17 from Elling's lectures states that in a regular Markov chain, the limiting distribution $\vec{\pi}$ gives the long-run fraction of time spent in eacg state. Thus, if we let N_{av} be the average number of days spent in each state over a year, $\frac{N_{av}}{365} = \pi_i \Rightarrow N_{av} = 365\pi_i$. We find $N_0 = 117.7$, $N_1 = 11.77$, $N_2 = 235.5$.

c)

See code file.

We let X be a stochastic variable representing the average number of days spent in a given state per year, averaged over the last 10 of 20 simulated years. We then compute 95% confidence intervals for the expected value of the average number of days spent in a state i per year, i.e. N_i. We now denote this as μ. By plotting a normal q-q plot for the long-run average of days spent in each state, see 2, we can see that the values are not too far from normally disrtributed. We therefore use the Central Limit Theorem to approximate X as normally distributed $N(\mu, \sigma^2)$. However, μ and σ are unknown, and we use the average over the 30 simulations of 20 years to estimate $\mu \approx \bar{x} = \frac{1}{n}\Sigma_{i=1}^{n} X_i$ and $\sigma \approx S = \sqrt{\frac{1}{n-1}\Sigma_{i=1}^{n}(X_i - \bar{X})^2}$. Since $X_1, ..., X_{30}$ can be viewed as normally distributed and independent, and μ, σ unknown, we can use the pivot $T = \frac{\bar{X}-\mu}{S/\sqrt{n}}$ which is student-t distributed with $\nu = n - 1 = 29$ degrees of freedom. We find the confidence interval, using quantiles $t_{0.025,29} = 2.045$:

$[\bar{X} - 2.045\frac{S}{\sqrt{n}}, \bar{X} + 2.045\frac{S}{\sqrt{n}}]$.

We simulate realisations $\bar{x}$ of $\bar{X}$ and realisations s of S, and get CIs

$[111.01, 134.19]$ for state S, $[10.53, 13.56]$ for state I, and $[220.59, 246.87]$.

Of course, these will vary between different realisations, but by computing the confidence intervals 1000 times and checking whether our analytical values were within the CI (see code file), we found that 95% of the time the analytical value π_i was within the computed interval. Thus, our computed intervals are in fact 95% confidence intervals.

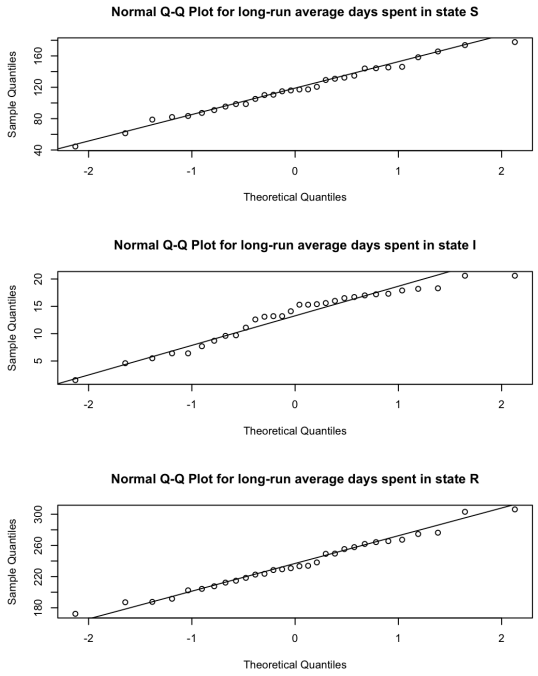


Figure 2: Normal Q-Q plot for average number of days spent in each state per year.

d)

Considering the sets $I_n : n = 0, 1, ...$, $Z_n : n = 0, 1, ...$ and $Y_n : n = 0, 1, ...$, where Z_n, Y_n denotes $Z_n = (S_n, I_n)$ and $Y_n = (S_n, I_n, R_n)$ respectively. We want to determine which ones fulfil the criteria for a Markov chain.

A Markov process is a stochastic process where the probability of any particular future behavior of the process, when its current state is known exactly, is not altered by additional knowledge concerning its past behavior [1].

$$\mathbb{E}[I_{n+1} \mid I_n = i,\ S_n = s]$$

$$\begin{cases} \text{if } \mathbb{E}[I_{n+1} \mid I_n = i,\ S_n = s] \text{ depends on S} \rightarrow \text{Not a Markov chain,} \\ \text{if } \mathbb{E}[I_{n+1} \mid I_n = i,\ S_n = s] \text{ independent of S} \rightarrow \text{is a Markov chain} \end{cases}$$

The main idea is that if we can show that the stochastic processes are only dependent on the states random variables at the previous timestep, then the stochastic process is a Markov chain.

The states in the next time step can be written as:

$$\begin{aligned} I_{n+1} &= I_n + I_{\text{new}} - R_{\text{new}}, \\ S_{n+1} &= S_n + S_{\text{new}} - I_{\text{new}}, \\ R_{n+1} &= R_n + R_{\text{new}} - S_{\text{new}}. \end{aligned}$$

Case 1:

$$I_{n+1} = I_n + \text{Bin}\left(S_n, \frac{0.5I_n}{N}\right) - \text{Bin}(I_n, \gamma).$$

$$\begin{aligned}\mathbb{E}[I_{n+1} \mid I_n, S_n] &= \mathbb{E}[I_n] + \mathbb{E}[I_{\text{new}}] - \mathbb{E}[R_{\text{new}}] \\ &= I_n + S_n\frac{0.5I_n}{N} - I_n\gamma.\end{aligned}$$

$\Rightarrow$ avhenger av S_n $\Rightarrow$ NOT Markov.

Case 2:

$$\begin{aligned}S_{n+1} &= S_n + S_{\text{new}} - I_{\text{new}} \\ &= S_n + \text{Bin}(R_n, \alpha) - \text{Bin}\left(S_n, \frac{0.5I_n}{N}\right).\end{aligned}$$

$$\begin{aligned}\mathbb{E}[S_{n+1} \mid S_n, I_n] &= \mathbb{E}[S_n] + R_n\alpha - S_n\frac{0.5I_n}{N} \\ &= S_n + (N - I_n - S_n)\alpha - S_n\frac{0.5I_n}{N}.\end{aligned}$$

$\Rightarrow$ depends only on S_n and I_n $\Rightarrow$ Z_n is a Markov chain.

Case 3:

$$\begin{aligned}R_{n+1} &= R_n + R_{\text{new}} - S_{\text{new}} \\ &= R_n + \text{Bin}(I_n, \gamma) - \text{Bin}(R_n, \alpha)).\end{aligned}$$

$$\mathbb{E}[R_{n+1} \mid R_n, S_n, I_n] = R_n + I_n\gamma - R_n\alpha.$$

$\Rightarrow$ depends only on R_n, S_n and I_n $\Rightarrow$ Y_n is a Markov chain.

e)

We let $Y_0 = (950, 50, 0)$ and simulate $n = 300$ steps of the Markov chain. See code file for exercise 1e.

$I_{new} \sim \text{Bin}(S_n, \beta_n)$, $S_{new} \sim \text{Bin}(R_n, \alpha)$, $R_{new} \sim \text{Bin}(I_n, \gamma)$.

We simulate, see code file, and get the time evolution for one selected realisation in 3.

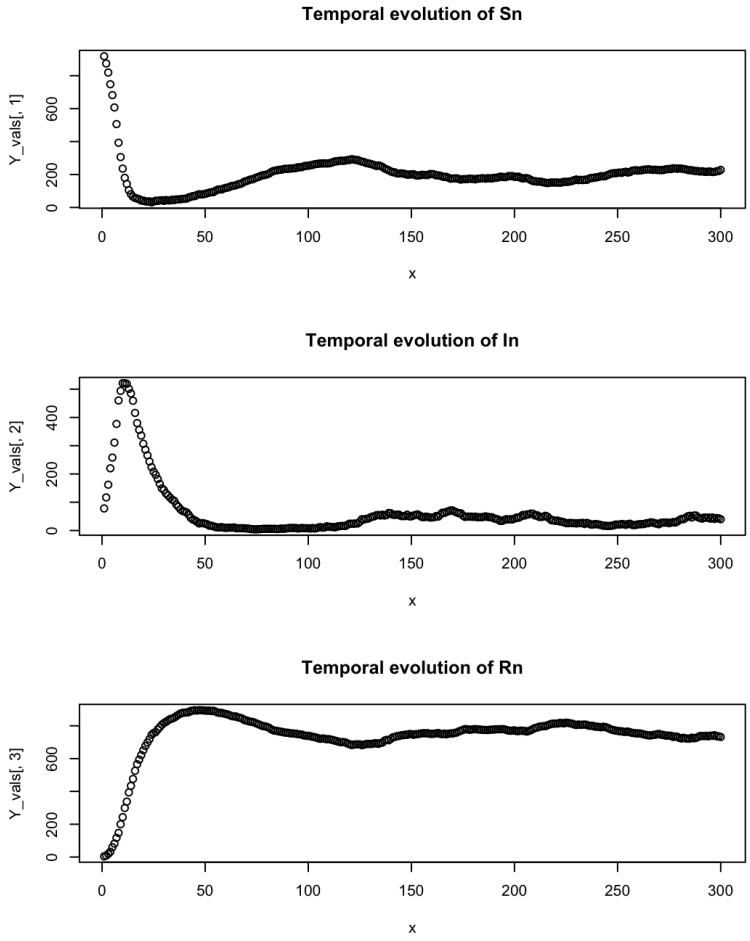


Figure 3: Temporal evolution of S_n, I_n and R_n.

We see from 3 a very different behaviour of the Markov chain in time intervals $[0, 50]$ and $[50, 300]$. Perhaps this can be explained intuitively as such: In the very first time steps, almost everyone is susceptible. We have some infected people, that can infect the susceptible people. As people are infected, β_n increases with I_n. However, the infected individuals become recovered with a probability of 0.1, and the recovered become susceptible again with a probability of only 0.005. We reach a "peak" where the maximum number of infeced individuals is reached. Then, the number of susceptible people left is very low, and therefore there are very few people to infect. As the infected population recovers, the infection rates remain low due to the low number of susceptibles and many immune people, and thus I_n decreases since more people are recovered than infected. After a while we therefore see the graphs somewhat flattening out and stabilizing at a certain, low number of infected individuals.

f)

Using the Central Limit Theorem again to approximate X as normally distributed $N(\mu, \sigma^2)$. Again both μ and σ are unknown, therefore we found the I_{max} and the time step where the first I_{max} occurs, for each 300 time steps, and approximated the expected value as the mean of these values over 1000 simulations. See the code file attached.

From the one of the simulations we obtained:

$\mathbb{E}[\max I_0, ..., I_{300}] \approx \frac{1}{300}\Sigma_{i=1}^{300} \max\{I_{i,1}, I_{i,2}, ..., I_{i,300}\} = \overline{I_{max}} = 523.5$, where $I_{i,k}$ is the number of infected individuals at time step k in simulation number i,

and

$\mathbb{E}[\min\{\arg\max\{I_0, ..., I_{300}\}\}] \approx \frac{1}{300}\Sigma_{i=1}^{300} \min\{\arg\max\{I_{i,1}, I_{i,2}, ..., I_{i,300}\}\} = \bar{T} = 11.189.$

Again, for the CIs, we get a pivot T that is student-t distributed and here the degrees of freedom are equal to $\nu = n - 1 = 999 \approx \infty$. The value of the quantile is found using the little blue book: $t_{0.025,\infty} = 1.960$. The CI for the expected values are then given by:

$[\overline{I_{max}} - 2.045\frac{S_I}{\sqrt{n}}, \overline{I_{max}} + 2.045\frac{S_I}{\sqrt{n}}]$ and $[\bar{T} - 2.045\frac{S_T}{\sqrt{n}}, \bar{T} + 2.045\frac{S_T}{\sqrt{n}}]$ respectively.

With values from one of the simulations of the expected values in the previous step, we obtain

$CI_I = [522.27, 524.83]$ and $CI_T = [11.821, 11.925]$.

So, with this method of computing the 95% CI the true value will lie somewhere in the computed interval with 95% probability. Therefore we can assess the severity of an outbreak by looking at how many people are likely to be infected at the most, and also how fast the disease will spread to infect this maximum number of individuals.

g)

We vaccinate $v = 100, 600, 800$ individuals, giving them life-long immunity, and see how this affects the temporal evolution of the disease within the population. We do so by simply subtracting v_i individuals from the susceptible population, while maintaining the population size $N = 1000$. If we were to reduce N by v, we could risk higher infection rates β_n as more people are vaccinated, which seems counterintuitive. See code files for code.

We run the modified simulation and obtain the following results:

Table 1: Expected values for I_{max} and T for $v = 0, 100, 600, 800$

v	$\overline{I_{max}}$	$\bar{T}$
0	522.374	11.864
100	439.584	12.69
600	97.627	15.77
800	49.792	2.022

We see from 1 that the maximum number of infected individuals $\overline{I_{max}}$ decreases quite

dramatically from $v = 100$ to $v = 600$. Also we notice that the time until the first max $\bar{T}$ increases somewhat. At $v = 800$ we can see that the number of infected individuals I_n in fact does not increase from its start value. The maximum is lower than the initial value, perhaps meaning the true maximum was obtained even before the first time step. This is a major improvement. The time until the first maximum is therefore lower again, since we never reach a "peak" as before, and I_n is strictly decreasing.

Below we see the temporal evoulutions for the different cases.

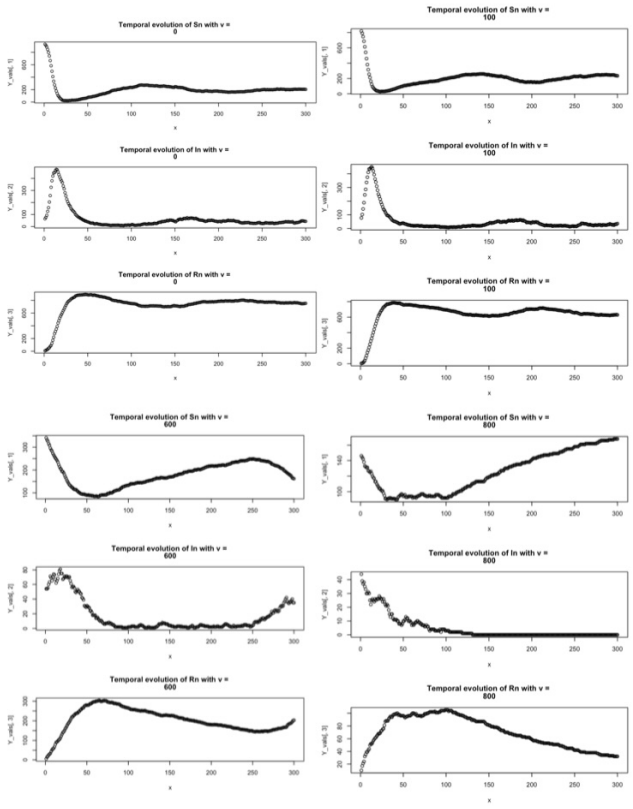


Figure 4: Temporal evolution of S_n, I_n and R_n for different numbers of vaccinated individuals v.

Problem 2

a)

For $X(t)$ number of claims received by an insurance company, where we assume $X(t) : t \geq 0$ can be modelled as a Poisson process, where t is measured in days since January 1st at $00 : 00.00$. We also have that the rate of the Poisson process is given by $\lambda(t) = 1.5, t \geq 0$.

For a homogeneous Poisson process, the probability mass function is given by

$$P(X(t) = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}, \qquad k = 0, 1, 2, \ldots \tag{1}$$

We want to calculate the probability of receiving more than 100 claims by March 1st at 00:00, after $t = 59$ days. Since $\lambda t = 1.5 \cdot 59 = 88.5$, we have

$$X(59) \sim \text{Poisson}(88.5). \tag{2}$$

Then the probability can be expressed as

$$P(X(59) > 100) = 1 - P(X(59) \leq 100) = 1 - \sum_{k=0}^{100} e^{-88.5} \frac{88.5^k}{k!}. \tag{3}$$

We use Python to calculate this numerically.

Listing 1: Exact probability of 100 claims after 59 days

```
t = 59
lam = 1.5 * t
claims = 100
sum = np.exp(-lam)
for i in range(1,claims+1):
    sum += np.exp(-lam)*(lam**i) / math.factorial(i)
probability_exact = 1-sum
print(probability_exact)
```

This gives an exact probability of approximately 0.103, meaning that there is about a 10.3% probability of receiving more than 100 claims during the first 59 days.

To verify this result, we simulate independent realizations of $X(59)$ using the Poisson distribution. The estimated probability is found by sorting out the simulations where the number of claims exceeds 100. For 1000 simulations we get an estimated probability of 0.105. This is close to the exact probabililty, but there will be deviations due to the randomness of the simulations.

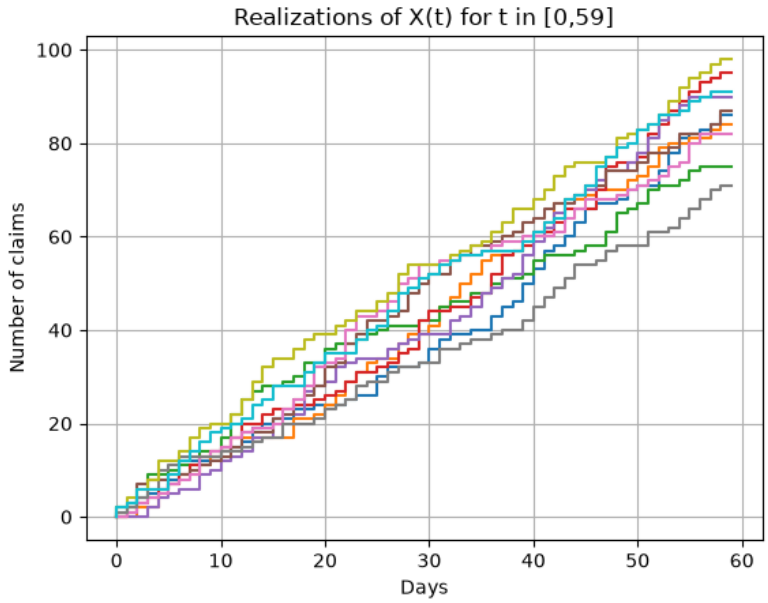


Figure 5: The plot shows how the number of claims increases over 59 days for 10 independant realizations of X(t).

Figure 5 shows ten realizations of the Poisson process, where the number of claims is increasing over time. However, the number of claims varies between realizations, again because of the randomness of the simulations. The expected number of claims at time t is $E[X(t)] = 1.5t$, giving an expected total of 88.5 claims after 59 days.

b)

We now consider the total monetary amount of claims received by the insurance company. We assume that each claim amount C_i follows an exponential distribution with rate parameter $\gamma = 10$ mill. kr, independently of the other claim amounts and of the claim arrival process.

The total claim amount at time t is defined as

$$Z(t) = \sum_{i=1}^{X(t)} C_i. \tag{4}$$

Since the claim amounts are measured in mill kr, the expected amount of an individual claim is

$$E[C_i] = \frac{1}{\gamma} = \frac{1}{10} = 0.1 \tag{5}$$

mill kr.

The process $Z(t)$ is a compound Poisson process, since both the number of claims and their monetary amounts are random.

To estimate the probability that the total claim amount exceeds 8 mill kr after 59 days, we simulate the number of claims using the Poisson distribution and generate an exponentially distributed monetary amount for each claim.

The estimated probability from the simulation is $p = 0.744$, found in the same way as in a, where we sorted out the simulations with a value above 8 mill kr.

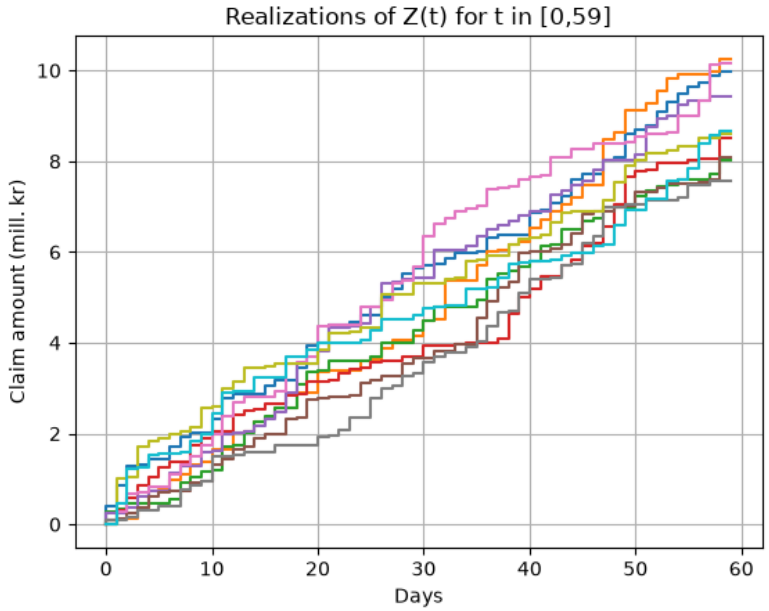


Figure 6: The plot shows how the total claim amount in mill. kr increases over 59 days, for 10 independant simulations of Z(t).

Figure 6 shows ten realizations of the cumulative monetary claim amount. The total amount increases with the new claims, where the size of the increase depends on the value of the corresponding claims. Therefore, the realizations deviate both in the number of claims and in the sizes of the increments.

The expected total claim amount after 59 days is

$$E[Z(59)] = E[X(59)]E[C_i] = 88.5 \cdot 0.1 = 8.85 \tag{6}$$

mill kr.

c)

We want to prove that $\{Y(t) : t \geq 0\}$ is a homogeneous Poisson process, where $Y(t)$ denotes the number of claims that exceed 250000 kr and need to be investigated.

We use the definition of a homogeneous Poisson process (Definition 1.1) from the interactive lecture notes, so we need to show that the increments are independent, that they are Poisson distributed, and that $Y(0) = 0$.

First, we find the probability that a claim exceeds 250000 kr. Since the claim amounts are measured in millions, this corresponds to 0.25 million. We know that $C_i \sim Exp(10)$, so

$$p = P(C_i > 0.25) = e^{-10 \cdot 0.25} = e^{-2.5}. \tag{7}$$

(1) Independent increments

Since $X(t)$ is a homogeneous Poisson process, the number of claims in disjoint time intervals are independent.

The claim amounts C_i are also independent of each other and of the arrival times. Therefore, the number of claims exceeding 250000 kr in disjoint time intervals are independent. Therefore, $Y(t)$ has independent increments.

(2) Poisson distributed increments

For any $s \geq 0$ and $t > 0$, we know that

$$X(s+t) - X(s) \sim Poisson(1.5t). \tag{8}$$

Each claim independently exceeds 250000 kr with probability $p = e^{-2.5}$.

Given that n claims arrive during the interval, the number exceeding the threshold follows a binomial distribution:

$$Y(s+t) - Y(s) \mid (X(s+t) - X(s) = n) \sim Binomial(n, p). \tag{9}$$

Since the number of claims is Poisson distributed and each claim independently exceeds the threshold with probability p, the resulting number is Poisson distributed with parameter $1.5pt$.

Therefore,

$$Y(s+t) - Y(s) \sim Poisson(1.5e^{-2.5}t). \tag{10}$$

(3) Initial condition

Since there are no claims yet at time $t = 0$, we have $X(0) = 0$. Therefore,

$$Y(0) = 0. \tag{11}$$

All three conditions in Definition 1.1 are satisfied. Therefore, $\{Y(t) : t \geq 0\}$ is a homogeneous Poisson process with rate

$$\lambda_Y = 1.5e^{-2.5} \approx 0.1231. \tag{12}$$

References

Pinsky, M.A, Karlin, S., *An Introduction to Stochastic Modeling.* Elsevier, 2011.